ACFI315 Principles of finance with Excel

Week 4 Statistical Analysis in Excel

A data file has been uploaded to VITAL.

1. Equity Index Data

Research questions:
(a) Calculate the weekly discretely compounded percentage returns for each of the data items.

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \frac{P_t}{P_{t-1}} - 1$$

(b) Find the average weekly return (**AVERAGE**).
(c) Find the deviation from the mean for each of the return series

$$Deviation = Return - Average$$

(d) Use conditional formatting to distinguish between values above the mean/below the mean (**SIGN**).

$$\begin{cases} 1\ if\ deviation > 0 \\ -1\ if\ deviation < 0 \end{cases}$$

(e) Find the deviation from the mean squared for each of the return series

$$Deviation^2 = (Return - Average)^2$$

(f) Find the sum of the deviations from the mean squared. If you divide this by N you have the variance of the population. Verify this result for each series using the **VAR.P** function. (**VAR.P**)

$$Var(Return) = \frac{1}{N}\sum_{i=1}^{N}(Deviation)^2$$

(g) Take the square root of the answer to part (f). This is the standard deviation. Verify this result for each series using the function **STDEV.P**. (**STDEV.P**)
(h) Use the **RANK** function to rank the series by mean return and standard deviation. Is there a noticeable pattern?
(i) Use the **SKEW** and **KURT** functions to find the skewness and kurtosis of the returns.
(j) Create a series of "bins" from -22% to 14% and use the **FREQUENCY** function to generate histogram data (**CTRL+SHIFT+ENTER**) for the returns.
(k) Use the **SIGN** function to find the weeks when the DAX and CAC moved together/apart. How many weeks did they move together/apart?
(l) for each week find $(DAX - Average) \times (CAC - Average)$. Find the sum of this, divide by N and find the covariance. Verify this result using the **COVAR** function.
(m) Divide your result in part (l) by the $standard\ deviation\ of\ DAX \times$ $standard\ deviation\ of\ CAC$. This is the correlation. Verify your result using the **CORREL** function.
(n) Use Data Analysis – Descriptive Statistics to verify the mean, variance, standard deviation, skewness and kurtosis of weekly returns.